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Clarifying social norms which have robustness against reputation costs and defector invasion in indirect reciprocity

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The evolution of cooperation through indirect reciprocity is a pivotal mechanism for sustaining large-scale societies. Because third parties return cooperative behaviour in indirect reciprocity, reputations that assess and share these third parties' behaviour play an essential role. Studies on indirect reciprocity have predominantly focused on the costs associated with cooperative behaviour, overlooking the costs tied to the mechanisms underpinning reputation sharing. Here, we explore the robustness of social norms necessary to secure the stability of indirect reciprocity, considering both the costs of reputation and the resilience against perfect defectors. Firstly, our results replicate that only eight social norms, known as the 'leading eight,' can establish a cooperative regime. Secondly, we reveal the robustness of these norms against reputation costs and perfect defectors. Our analysis identifies four norms that exhibit resilience in the presence of defectors due to their neutral stance on justified defection and another four that demonstrate robustness against reputation costs through their negative evaluation of unjustified cooperation. The study underscores the need to further research how reputational information is shared within societies to promote cooperation in diverse and complex environments.

Keywords Evolution of cooperation, Indirect reciprocity, Social norm, Agent-based simulation

Among the several rules that bring about cooperation[1], indirect reciprocity[2] is a powerful mechanism for maintaining cooperation in large-scale societies that only humanity has acquired. Although several non-human species can achieve direct reciprocity[3,4], indirect reciprocity is unique to humans and humans are known to acquire the capacity for indirect reciprocity in early childhood[5]. To make indirect reciprocity work, individuals need a high capacity to remember the behaviour of other members of society and a social foundation for sharing each member's behaviour. The key to this foundation is reputation, where people select who they should cooperate with by sharing the behaviour of third parties as a reputation. In a real human society, literature has revealed that people spend much time sharing the reputations of others as gossip[6,7] and that people are willing to pay a great price to maintain their reputation[8].

A vast amount of literature has looked for the conditions under which indirect reciprocity can stabilise cooperation. Indirect reciprocity is characterised by cooperative behaviour being returned by third parties rather than the original recipient, as is the case with direct reciprocity. Therefore, determining cooperative behaviour requires information about others who may be potential future recipients. There needs to be a mechanism whereby benefits are returned to those who pay the cost of cooperation by cooperating only with good persons, using assessment rules that distinguish between good and bad persons. In the framework of indirect reciprocity studies, social norms are regarded as the assessment rules that distinguish between good and bad and are shared by everyone in society. Early studies proposed a simple social norm: 'cooperation is good and defection is bad'[9,10]. However, this norm has an Achilles heel in that once a non-cooperator invades (or an error causes non-cooperation), cooperation collapses as non-cooperation cascades. Social norms that can overcome this Achilles heel include 'cooperating with a bad person is bad and defecting against a bad person is good'[11,12], and

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'cooperation even with a bad person is good and defecting against a bad person is also good'[13–15]. Eight social norms, known as the 'leading eight', can maintain cooperation stably and form one of the foundations for research on indirect reciprocity[16–18]. However, these studies have all dealt with the costs of the cooperative behaviour itself but have yet to consider the costs of the reputation-sharing mechanisms that underpin cooperation.

Different forms of reciprocity in cooperative behaviour entail various constraints and requirements. For example, for kin selection[19], there is a strong constraint that only genetically close partners can co-operate. Direct reciprocity requires the assumption that individuals interact repeatedly with each other[20,21]. Indirect reciprocity is free from these constraints, but it should be noted that it comes at the cost of maintaining a basis for diffusing and sharing reputational information. Little literature has focused on the costs of reputation in indirect reciprocity[22–24].

Other perspectives on reputation need to be considered besides the cost of sharing reputation. There are two main approaches to treating reputational information in indirect reciprocity. One is 'public information'[9,10,16], where all members share a reputation, and the other is 'private information'[25–27], where individual members hold a reputation uniquely. This paper explores social norms robust to reputation costs in a public information situation.

Most literature analyses whether a social norm can sustain cooperation on the basis of whether it is 'evolutionally stable.' In this context, the term 'stable' means that the corresponding social norm can prevent invasion from other action rules[16,17] or unconditional strategies such as perfect cooperation and perfect defection[28,29]. However, the question of whether cooperation can spread and a cooperative society can be achieved from a chaotic society where social norms are not well shared is a significant challenge in the evolution of cooperation. Two possible approaches exist to analyse whether cooperation can evolve from a non-cooperative regime. One is to analyse whether the focal social norm can invade a population dominated by perfect defectors[29,30], and the other is to analyse whether the focal social norm is selected within a diverse array of social norms[31,32]. We focus on the intrusiveness of social norms on perfect defectors. Because perfect defectors are the simplest yet most powerful destroyer of cooperation, analysing robustness against them has important implications for realising a cooperative society.

On the basis of a representative model[16,17] that exhaustively explores social norms of indirect reciprocity, this paper explores robust and stable social norms concerning both reputation costs and invasions by perfect defectors. We employ an agent-based simulation (ABS) to understand the model's behaviour in a wide parameter space. After reproducing the stability of the 'leading eight' norms with ABS, we identify norms that are robust and stable against reputation costs and perfect defectors.

Model

Here, we define a model of indirect reciprocity with finite $N < \infty$ individuals who play a giving game with no spatial structures. In the giving game, firstly a pair of a donor and a recipient is randomly chosen from the population. Secondly, the donor decides whether to cooperate with the recipient by paying a cooperation cost $c > 0$. If the donor chooses to cooperate, the recipient receives a benefit $b > c$; otherwise, the recipient obtains nothing. The donor chooses cooperation (hereafter, denoted C) with or defection (D) against a recipient using one's own action strategy defined as an action function.

A unique and public reputation system gives binary reputation labels to players where $I_i(t) \in \{\mathrm{good}, \mathrm{bad}\}$ is defined as a reputation label for Player i at simulation time t. All players use the labels provided by this system when deciding one's own action towards a potential recipient.

An assessment rule is denoted as a four-letter string (for example, [GGBB]). The meanings of each letter are as follows. The first letter is a response for [C to good], a response for the case when the donor playing a giving game gives C to a potential recipient with a good label. The second letter is a response for [C to bad] where the donor gives C to the recipient with a bad label. The third letter is for [D to good] while the fourth letter is for [D to bad]. Following a previous study[33], the set of responses is denoted as $S = \{\mathrm{G},\mathrm{B},\mathrm{K},\mathrm{R}\}$, the symbols of which indicate 'Assign good' , 'Assign bad', 'Keep the label unchanged', and 'Reverse the label between good and bad', respectively. For example, if a society adopts a social norm as an assessment rule [GGBB], whoever cooperates with someone in the giving game should be labelled as good while whoever defects against someone should be labeled as bad. This norm corresponds to the so-called Image-Scoring norm[9,10].

The donor chooses cooperation with or defection against a recipient using his/her action strategy. A combination of the donor's reputation and the recipient's reputation describes action strategies, resulting in 16 different strategies (Table 1). The action strategy is represented by a 4-bit code, e.g. CCDD cooperates if the

	Reputation of the recipient	
	Good	Bad
Reputation of the donor		
Good	C/D	C/D
Bad	C/D	C/D

Table 1. Action strategy of donor: Donors use their own reputations and the reputations of their recipients to determine their behaviour. There are 16 action strategies, ranging from DDDD to CCCC.

donor's reputation is good and defects if it is bad, while CDCD cooperates if the recipient's reputation is good, and defects if it is bad.

Using our notation, the total number of possible assesment functions is $256 = 4^4$, and action strategies is $16 = 2^4$. By combining them, we obtain the classical third-order norms studied before[16].

When an agent acts as a donor, all but the perfect defector (DDDD) and perfect cooperator (CCCC) pay $\gamma(> 0)$ as a cost to access reputation information. The donor's behaviour is evaluated and judged as good or bad in accordance with the social norm, and all members share the result. All agents employ the same social norm. Figure 1 illustrates the donor's choice of action, the update of payoffs by the giving game, and the update of donor reputation by a social norm.

The distribution of action strategies at the start of the simulation is that there are DDDD at population rate η and CDCD, which cooperate with recipients of good reputation with population rate $1 - \eta$. The simulation runs for *G* generations, and within one generation, each member plays the giving game as donor and recipient *T* times. At the end of each generation, the agents update their strategies. We define the cumulative payoff of agent *i* as U_i. Agent *i* imitates agent *j*'s strategy with probability P_{ij} compared to *j* randomly chosen from the population and does not change its strategy with probability $1 - P_{ij}$. The $s(> 0)$ is the temperature parameter.

$$P_{ij} = 1/[1 + \exp(-s(U_j - U_i))] \tag{1}$$

We introduce two perturbation factors: errors and mutations. When an agent acts as a donor, an implementation error occurs with probability e_1 and its behaviour (C/D) is inverted. Also, when social norms are used to assess an agent's behaviour, an assessment error occurs with probability e_2, and its reputation (good/bad) is inverted. When updating the strategy, a mutation occurs with probability *m*, and the agent's action strategy is set to a randomly selected one from 16 different strategies.

We define the average cooperation ratio of the last G' generations as the average cooperation ratio of the simulation. We set the values of the fixed parameters as follows. $N = 200, T = 500, G = 200, G' = 100, c = 1, b = 2, s = 1, e_1 = e_2 = m = 0.02$ We set *N*, *T*, *G* and G' to ensure that the group size is sufficient to guarantee diversity and that there is sufficient time for behaviour to stabilise. We also adopted the values of other parameters from the previous pioneering paper[16].

Results

We firstly ran simulations under the condition $\gamma = \eta = 0$ to check consistency with the results of previous studies[16,17]. The simulation results showed that only eight social norms, the leading eight, achieved a cooperation ratio of over 0.9 (see the Supplementary Data). The action strategy CDCC was dominant only for social norms GGBK (L1) and GRBK (L2), while CDCD was dominant for the other social norms. This result is also consistent with previous studies.

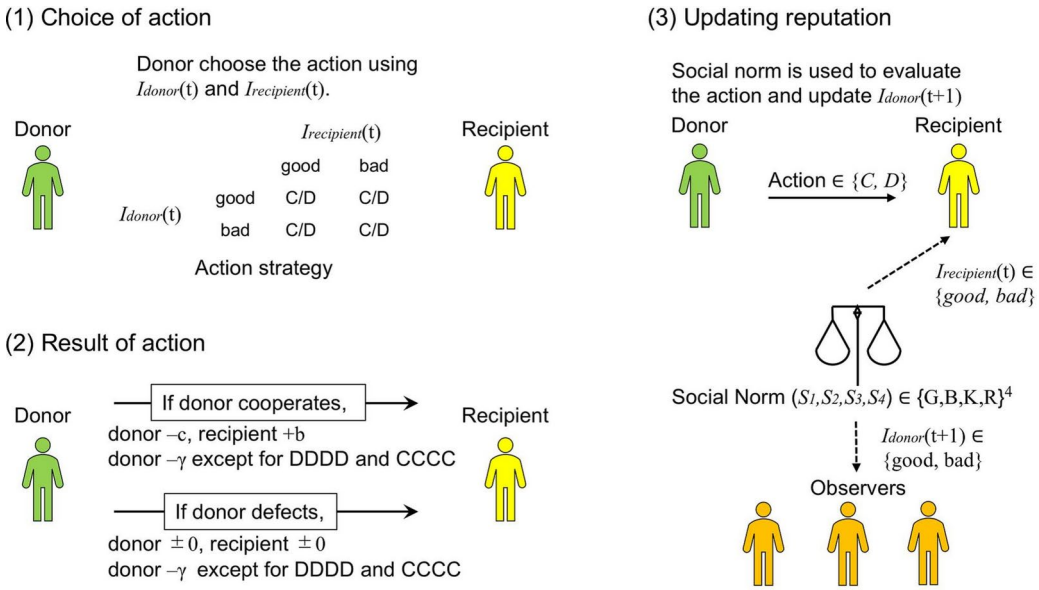


Figure 1. An outline illustration of the model: (1) A donor and recipient selected randomly from the population play the giving game. The donor chooses whether to cooperate or not in accordance with an action strategy. The behavioural strategy is defined in Table 1. (2) When the donor cooperates, he/she pays *c*, and the recipient receives *b*. When the donor does not cooperate, the payoffs of both remain unchanged. Donors other than perfect cooperators and perfect defectors pay a reputation cost γ regardless of their behaviour. (3) The donor's behaviour is evaluated in accordance with the social norm, and the donor's reputation is updated. All members share a specific social norm.

We then explored social norms that are robust to γ and η. Figure 2 shows the cooperation ratio for various values of γ and η for each of the 'leading eight' norms. Blue in the heatmap indicates cooperation rate = 1, while red corresponds to cooperation rate = 0. We repeated 20 simulation trials for each γ, η combination using different random seeds and adopted the average value as the cooperation ratio. Figure 2 makes clear that the social norm GBBK (L8) has the most widest range of the parameter space to achieve cooperation. This norm is characterised by an unchanged reputation for defection against a recipient with a bad reputation (justified defection), and a rating of cooperation with a recipient with a bad reputation (unjustified cooperation) as bad.

The simulation results reveal two noteworthy phenomena. The first notable result is the comparison of the top row's four panels with the bottom row's four panels in Fig. 2. The two panels above and below are paired in each column; the difference between them is that the fourth bit of the norm is 'G' or 'K'. The fourth bit is how justified defection is evaluated. In all four pairs, the parameter range for η in which cooperation can evolve is larger when the evaluation is unchanged (K) than when justified defection is justified (G).

The second feature is robustness to reputation cost (γ). A feature of robust norms against reputation costs is that the second bit is (B or R). The second bit represents [C to bad], that is how unjustified cooperation is evaluated. B/R of the second bit means that unjustified cooperation is evaluated as bad. Because 'R' means reversing reputation, in a cooperative regime society, unjustified cooperation is judged as bad because many agents have a good reputation. In other words, norms that judge unjustified cooperation as bad maintain cooperation robustly against reputation costs. The reason is that these norms can exclude the perfect cooperator. The perfect cooperator can be regarded as a free-rider to reputation. Integrating these results, we found that GBBK (L8) can simultaneously eliminate the first-order free-rider (perfect defector) and the second-order free rider (perfect cooperator) and achieve cooperation over a wide area.

Figure 3 shows the time evolution of the average cooperation ratio of the population and the population ratio of each action strategy. The four norms in the bottom row are vulnerable to perfect defectors, so cooperation collapses quickly. Symmetrically, the four norms shown in the upper row can invade into perfect defectors, and thus, cooperation temporarily evolves. However, in L1 and L7, cooperation eventually collapses. Since these two norms do not negatively assess 'unjustified cooperation,' they are susceptible to invasion by perfect cooperators who also cooperate with bad actors. In societies with a cooperative regime, perfect cooperators consistently hold an advantage actors using over other strategies because they incur no reputation costs. Consequently, perfect cooperators proliferate within the group and become predominant. This expansion ultimately leads to an invasion by perfect defectors, resulting in the collapse of cooperation. On the other hand, L2 and L8 do not cooperate with perfect cooperators because they assess unjustified cooperation negatively. Consequently, perfect cooperators fail to infiltrate the population, ensuring that cooperation remains robust.

As Fig. 2 shows, a white-coloured region with a medium cooperation rate lies at the boundary between the cooperative and non-cooperative regimes. To analyse the phenomena observed in this region, we focus on the social norm GBBK (L8) and show in Fig. 4 the time evolution of the behavioural strategies according to the value of γ. Panel (A) depicts the case where γ is small, cooperation is stable, and the behavioural strategy CDCD is dominant. Panel (D) illustrates the case with large γ, where cooperation collapses and the behavioural strategy DDDD is dominant. However, panel (B) shows peculiar behaviours, such as cyclic dominance between CCCC and CDCD It also shows the occasional intrusion of DDDD into that cycle. Surprisingly, this suggests the existence of a cooperation regime with a dynamic cycle between a stable CDCD-dominated cooperation state

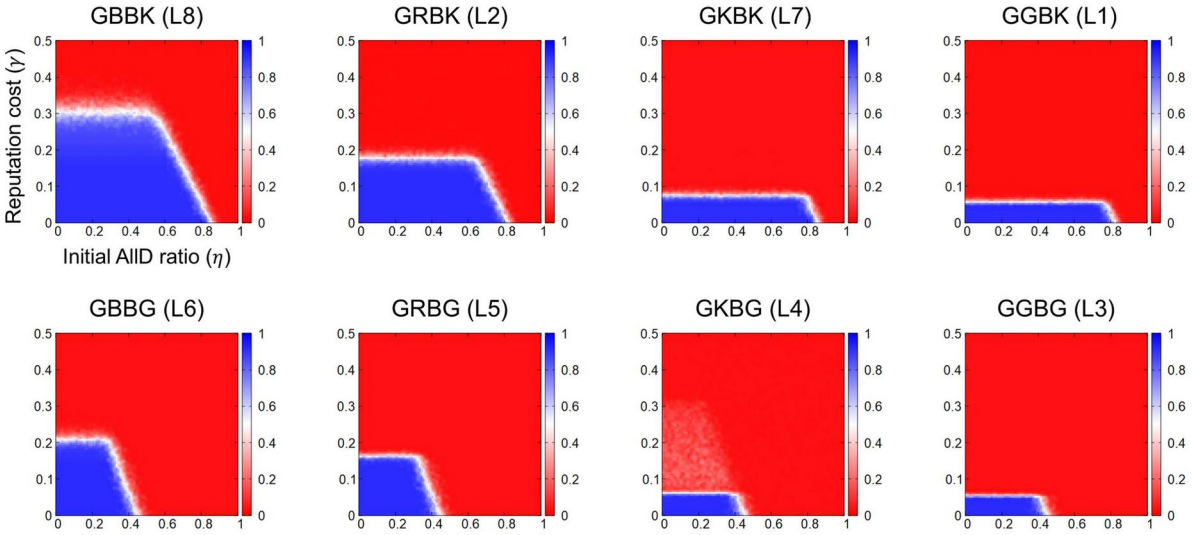


Figure 2. Phase diagrams of cooperation ratios in population: x-axis represents the initial perfect defector ratio (η), and y-axis represents reputation cost (γ). In accordance with the values of η and γ, there are two phases: one in which cooperation dominates (shown in blue), and another in which defection dominates (shown in red). The four norms in the upper row achieve cooperation robustly for η, while the four on the left side achieve cooperation robustly for γ. The social norm GBBK (L8) has the widest cooperation area.

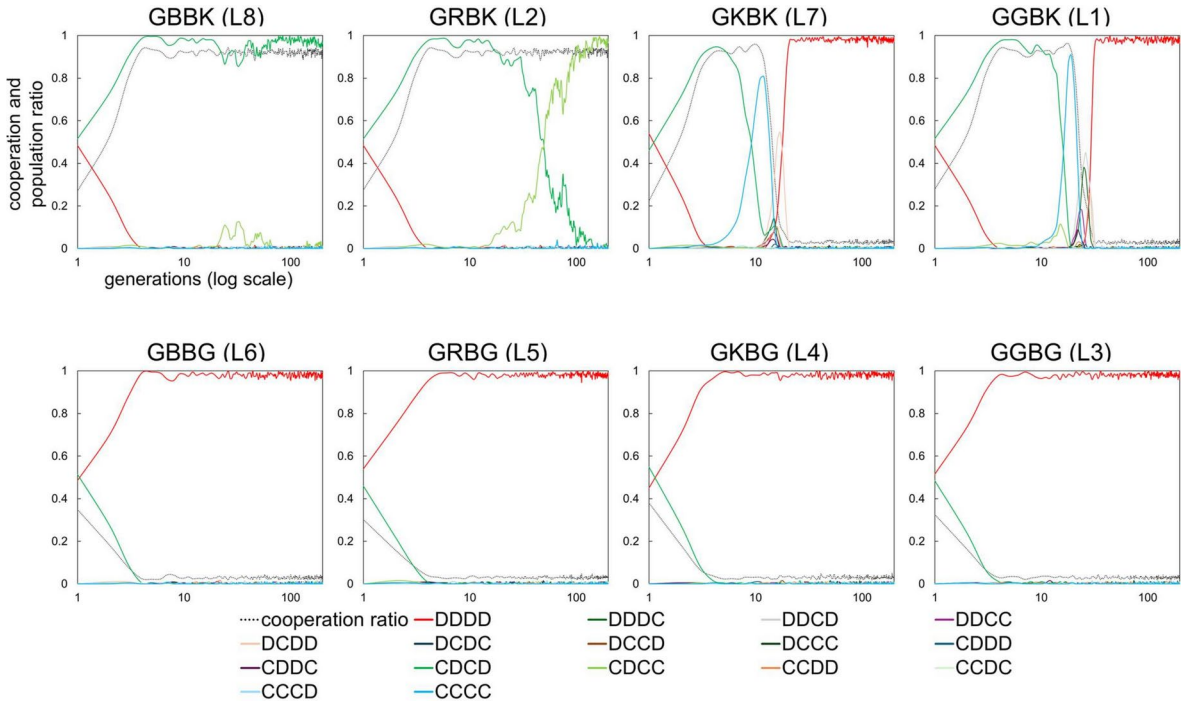


Figure 3. A typical pattern of time evolution of cooperation and action strategies: Each panel shows a typical run of simulations with reputation costs and the initial ALLD population ($\eta = 0.5$ and $\gamma = 0.1$) x-axis represents the generations (log scale), and y-axis represents cooperation and population ratio. The dotted black line shows the cooperation ratio, and the solid coloured lines show the population ratio of each action strategy. Only GBBK (L8) and GRBK (L2) can maintain cooperation because they can invade into perfect defectors and never permit the invasion by perfect cooperators.

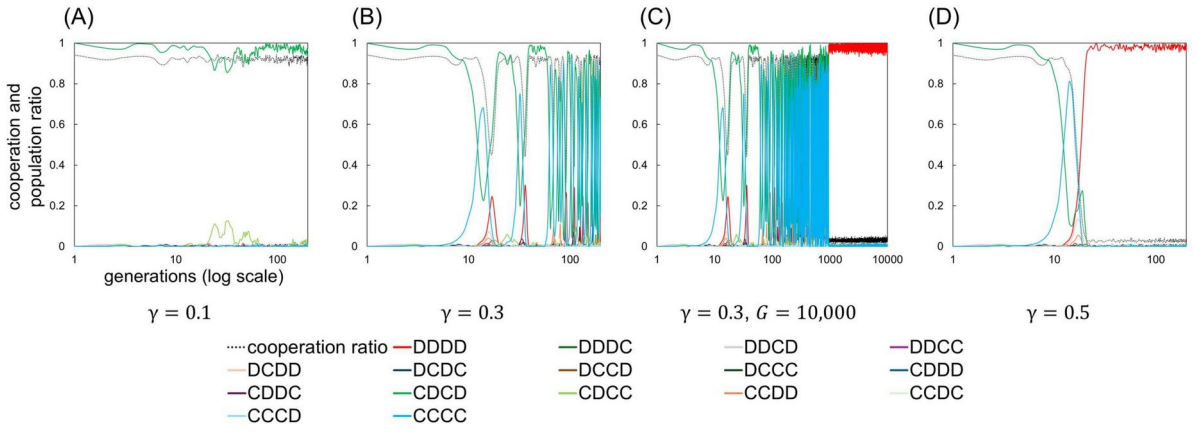


Figure 4. A typical pattern of time evolution of cooperation and action strategies in accordance with the values of γ: Each panel shows a typical run of simulations with various reputation costs ($\eta = 0.0$ and $\gamma = 0.1, 0.3$ and 0.5, respectively) x-axis represents the generations (log scale), and y-axis represents cooperation and population ratio. In the case of panel (C), generations G is set to 10, 000; in other panels, $G = 200$. The dotted black line shows the cooperation ratio, and the solid coloured lines show the population ratio of each action strategy. In all panels, the social norm is GBBK (L8).

and a stable DDDD-dominated non-cooperation state. However, when a tremendous value is set for generation G, it is found to be sucked into DDDD dominance due to the effect of randomness, as shown in panel (C). Although the complexity of the simulation model, with its mixture of 16 different behavioural strategies, makes it difficult to demonstrate cycle stability, a promising future development is to explore cycle stability through more concise mathematical models.

Discussion

This paper investigates the impact of reputation costs on the evolution of cooperation through indirect reciprocity employing ABS. Firstly, our simulations replicated previous studies[16,17], verifying that only eight social norms, called the leading eight, successfully establish a cooperation regime. We then analysed whether the leading eight norms are robust to reputation costs and perfect defectors. The results show that four norms are robust in the presence of defectors. These four are distinguished by their neutral stance on justified defection, neither evaluating it good nor bad while maintaining their evaluations. Additionally, from a different perspective, four norms demonstrated robust properties against reputation costs. A defining feature of these norms is their negative evaluation of unjustified cooperation. In sum, unjustified cooperation should not be evaluated tolerantly, and evaluation should not be changed by justified defection.

Remarkably, the four norms that are robust for reputation costs (shown in the left half of Fig. 2) correspond to those that fail to maintain cooperation in private information environments[33]. Put differently, norms capable of sustaining cooperation in environments with private information are susceptible to reputation costs. Considering that private information environments lack institutionalised reputation systems and consequently do not bear reputation costs, the leading eight can be categorised into two distinct classes. The first group consists of four norms suited for public information environments, capable of sustaining robust cooperation despite reputation costs. The second group includes four norms that are suitable for private information environments due to their susceptibility to reputation costs. The difference between the two lies in the former (B/R) rendering a negative judgment on unjustified cooperation, while the latter (G/K) assigns a positive one.

Previous research has paid little attention to reputation costs in indirect reciprocity. However, maintaining public reputation systems in the real world is costly[34] (e.g., Trip Advisor word-of-mouth or Amazon reviews). We also devote considerable time and effort to coordinating and maintaining private reputation systems, such as gossip[6]. The relationship between reputation costs and the type of reputation system, i.e. public or private, remains largely unexplored. The previous papers[22,24] allowed for a second-order free rider problem because the discriminators can also free ride on the reputation. In our model, the discriminator consistently incurs a reputation cost. This approach allows the model to identify and reveal robust social norms against reputation costs. The experimental study[23] employed a setting that incurs costs to obtain second-order information (recipient's reputation). Future research should focus on varying costs on the basis of the type of reputation. Some studies also mentioned institutional sanctions[24], costs of obtaining additional reputation information[35,36], and the costs of complex indirect reciprocity strategies[37]. More realistic and sophisticated modelling of reputation costs should be explored in future works. Our findings indicate a need to further investigate how societies and individuals must share reputational information to foster cooperation in varied and complex environments.

Several studies highlight that maintaining a neutral evaluation for justified defection penetrates more effectively into populations of perfect defectors. Our findings reveal that all four norms that retain their evaluations for justified defection are intrusive into perfect defectors[27,38]. Moreover, evidence from an experimental study suggests that individuals in actual societies maintain neutral evaluations and reserve their judgments concerning justified defection[39]. Justified defection can be interpreted as a punitive action against first-order free riders (perfect defectors) in indirect reciprocity. Our results suggest that avoiding active endorsement of punishment plays a crucial role in the evolution of cooperation. Several studies have noted that individuals who carry out punishments are not positively evaluated[40–42].

To summarise the discussion, we can organise the leading eight as depicted in Table 2. The norms 'GRBK (L2)' and 'GBBK (L8)' demonstrate robustness to both reputation costs and perfect defection. A comprehensive analysis of the leading eight pointed out[27] that 'GRBK (L2)' can compete with ALLC and ALLD, but no paper has shown the distinguishing features of 'GBBK (L8)'[18,43]. However our results revealed that these two norms demonstrate robustness against both reputation costs and perfect defection. Considering the extent of the parameter space where cooperation is achieved, 'GBBK (L8)' stands out as the leader among the leading eight in these two dimensions.

It is important to recognise several limitations in our study. Firstly, this paper does not provide a mathematically rigorous solution for the conditions that settle cooperation, as it utilises ABS. A mathematical analysis would clarify the conditions necessary for achieving cooperation. Specifically, conducting a mathematically rigorous exploration of the conditions that allow for observing of dynamic equilibrium due to cyclic behaviour represents a crucial topic for future research. Secondly, we must check effects of benefit-to-cost ratios (b/c) on cooperative social norms. In the paper, we examined whether cooperation could be achieved at relatively low benefit-to-cost ratios ($b/c = 2$). Recent studies have shown that other norms can also achieve cooperation in the $b/c > 2$ regions[44]. We expect that the relationship between reputation costs (γ) and the benefit-to-cost ratio (b/c) plays a crucial role in determining the conditions under which cooperation can be sustained. Future research needs to focus on rigorously identifying the maximum threshold of reputation cost and cooperation efficiency necessary

	Resilience for reputation costs	
	Strong	**Weak**
Resilience from free-riders		
Strong	GBBK (L8), GRBK (L2)	GKBK (L7), GGBK (L1)
Weak	GBBG (L6), GRBG (L5)	GKBG (L4), GGBG (L3)

Table 2. The summary of simulation results.: The leading eight are divided into four in accordance with two resilience indices.

for cooperation to emerge. Finally, connecting these findings with real-world scenarios poses a future challenge. Experimental studies are needed to determine whether people in public information environments actually adopt the most effective norms. Our results also prompt further questions about how responses to people with negative reputations should be evaluated. Justified defection and unjustified cooperation represent punishment and the slacking off of punishment efforts, respectively. Future research should combine theoretical and experimental investigations to elucidate the still unclear role of punishment in fostering cooperation.

Data availibility

A part of data of this study are stored in an OSF data package titled 'Data of Clarifying social norms which have robustness against reputation costs and defector invasion in indirect reciprocity', which can be accessed at the below link. Other data is available from the corresponding author on reasonable request. https://doi.org/10.17605/OSF.IO/R7PHF

Received: 8 August 2024; Accepted: 10 October 2024
Published online: 23 October 2024

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Author contributions

H.Y. initiated and performed the project; H.Y. implemented and executed the simulation; H.Y., S.U. and T.S. analysed the data; H.Y. and I.O wrote the paper; All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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